

Chenglong Fu

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RESEARCH INTERESTS

My research centers on securing emerging technologies through three interconnected pillars: IoT and cyber-physical system security, AI-enhanced security frameworks, and the development of trustworthy AI. Additionally, I have a strong interest in exploring the deployment and application of generative artificial intelligence in disciplinary research.

EDUCATION

Temple University

Ph.D. in Computer and Information Sciences

Philadelphia, PA

Aug. 2016 – May. 2022

- Supervised by [Prof. Xiaojiang Du](#)
- Dissertation Title: Semantics-aware Effective Security Solutions for Home Automation System

University of Science and Technology of China

Bachelor of Engineering in Information Security

Hefei, China

Aug. 2011 – June. 2015

RESEARCH & WORK EXPERIENCES

Assistant Professor

University of North Carolina at Charlotte

Aug 2022 – Present

Charlotte, NC

Postdoc Research Fellow

University of South Carolina

Jan. 2022 – Aug. 2022

Columbia, SC

Research Assistant

Temple University

Sep. 2016 – December. 2021

Philadelphia, PA

SELECTED PEER REVIEWED PUBLICATIONS

- [1] ZhuoYi Shi, **Chenglong Fu**, Xiaojiang Du, Rong Xu. “Effective Handling of Erroneous Executions of Automation Rules on Multiple Platforms Smart Home IoT Systems”. (under review)
- [2] Rong Xu, **Chenglong Fu**, Xiaojiang Du, ZhuoYi Shi, Hao Wang, Nicholas Bielanos. “EdgeFuzzer: Context-guided Fuzzing for IoT Applications on Smart Home Hubs”. (under review)
- [3] Huan Bui, **Chenglong Fu**, Fei Zuo. “TAPAssure: Safe Automation in Smart Homes with LLMs and Formal Verification”. Accepted to EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (SmartSP 2025), December, 2025
- [4] Danial Abshari, Peiran Shi, **Chenglong Fu** (corresponding author), Meera Sridhar, Xiaojiang Du. “INVARLLM: LLM-assisted Physical Invariant Extraction for Cyber-Physical Systems Anomaly Detection”, (under review)
- [5] Andrew D’Angelo, **Chenglong Fu** (corresponding author), Xiaojiang Du. “Phantom Buster: Defense against the Stealthy IoT Delay Attack in Smart Homes”, submitted to IEEE International Conference on Computer Communications (INFOCOM) 2026. (under review).
- [6] Xuening Xu, **Chenglong Fu**, Xiaojiang Du, Bo Luo. “EdgeDevil: The Dormant Menace in Your Smart Home”, (under review).
- [7] Xuening Xu, **Chenglong Fu**, Xiaojiang Du, Bo Luo. “IoTtracer: Detecting Malicious Local Attacks on IoT Devices by Utilizing IoT System-level Traffic Patterns”, (under review).

- [8] Fei Zuo, Junghwan Rhee, Yung Ryn Choe, **Chenglong Fu**, Xianshan Qu. “Few-Shot Learning-Based Cyber Incident Detection with Augmented Context Intelligence”. Accepted by IEEE Annual International Computer Software and Applications Conference (COMPSAC) 2025.
- [9] Xuening Xu, **Chenglong Fu**, Xiaojiang Du, Bo Luo. “Discovering and Exploiting IoT Device Hidden Attributes: A New Vulnerability in Smart Homes”, Accepted by ACM Conference on Computer and Communications Security (CCS) 2025.
- [10] Huan Bui, Harper Lienerth, **Chenglong Fu**, and Meera Sridhar. “Poster: TAPChecker: Model Checking in Trigger-Action Rules Generation Using Large Language Models.” Poster presented at the ACM Conference on Computer and Communications Security (CCS) 2024, Salt Lake City, Utah, October 2024.
- [11] Haotian Chi, Qi Ma, Yuxuan Zhang, **Chenglong Fu**, Yuwei Wang, Haijun Geng, and Xiaojiang Du. “Audio-Assisted Smart Home Security Monitoring with Few Samples.” In Proceedings of the 2024 IEEE Global Communications Conference: Communication & Information Systems Security (Globecom 2024 CISS), Cape Town, South Africa, December 7, 2024, pp. 1–6.
- [12] Huan Bui, **Chenglong Fu**, “ShadowConn: Breaking the Entanglement of Cross-platform IoT Delegation in Multi-user Environments”, Accepted by EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles 2024.
- [13] Md Rabbi Alam, Jinpeng Wei, Md Sajidul Islam Sajid, Qingyang Wang and **Chenglong Fu**, “Attacking IoT Devices through the IoT Cloud Platforms: An Empirical Study”, Accepted by ACM Conference on Data and Application Security and Privacy (CODASPY) 2024
- [14] Trent Reichenbach, **Chenglong Fu**, Xiaojiang Du, Jia Di, Yuede Ji, “TrustEvent: Cross-Platform IoT Trigger Event Verification Using Edge Computing”, Accepted by IEEE International Conference on Communications (ICC), June 2024.
- [15] **Chenglong Fu**, Xiaojiang Du, Qiang Zeng, Zhenyu Zhao, Fei Zuo, Jia Di, “Seeing Is Believing: Extracting Semantic Information from Video for Verifying IoT Events”, Accepted by 17th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec), May 2024.
- [16] Andrew D’Angelo, **Chenglong Fu**, Xiaojiang Du, E. Paul Ratazzi. “Discovering Complex Correlations among Multiple IoT Devices in Smart Environments”, IEEE Global Communications Conference (Globecom), December 2023.
- [17] Andrew F D’Angelo, Chenglong Fu, Xiaojiang Du, Paul Ratazzi, “Transformer-based Compound Correlation Miner for Smart Home Anomaly Detection”, IEEE International Conference on Cloud Networking (CloudNet), November 2023.
- [18] Yunping Fang, **Chenglong Fu**, Xiaojiang Du, “Virtual-Device-Based Policy Enforcement in Multi-Admin Smart Environments”, IEEE International Conference on Cloud Networking (CloudNet), November 2023.
- [19] Xuening Xu, **Chenglong Fu**, Xiaojiang Du. “MP-Mediator: Detecting and Handling the New Stealthy Delay Attacks on IoT Events and Commands”, 25th International Symposium on Research in Attacks, Intrusions and Defenses (RAID) November 2023. (Acceptance rate: 23.5%)
- [20] Xuening Xu, **Chenglong Fu**, Xiaojiang Du, and E. Paul Ratazzi. “VoiceGuard: An Effective and Practical Approach for Detecting and Blocking Unauthorized Voice Commands to Smart Speakers”, IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), June 2023. (Acceptance rate: 19.6%)

- [21] Chenxu Jiang, **Chenglong Fu**, Zhenyu Zhao, Xiaojiang Du, “Effective Anomaly Detection in Smart Home by Integrating Event Time Intervals”, International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN), October 2022.
- [22] **Chenglong Fu**, Qiang Zeng, Xiaojiang Du, Siva Likitha Valluru. “Demystifying and Exploiting IoT Timeout Behaviors in Smart Homes”, IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), June 2022. (Acceptance rate: 18.7%)
- [23] Haotian Chi, **Chenglong Fu** (Co-first author), Qiang Zeng, Xiaojiang Du. “Delay Wreaks Havoc on Your Smart Home”, Accepted by IEEE Symposium on Security and Privacy (IEEE S&P), May 2022. (Acceptance rate: 15.2%)
- [24] Xuanyu Liu, Qiang Zeng, Xiaojiang Du, Siva Likitha Valluru, **Chenglong Fu**, Xiao Fu, Bin Luo “SniffMislead: Non-Intrusive Privacy Protection Against Wireless Packet Sniffers in Smart Homes.” In the International Symposium on Research in Attacks, Intrusions and Defenses (RAID), October 2021. (Acceptance rate: 23.9%)
- [25] **Chenglong Fu**, Qiang Zeng, Xiaojiang Du. “HAWatcher: Semantics-Aware Anomaly Detection for Appified Smart Homes. ” In the proceeding of 30th USENIX Security Symposium (USENIX Security), August 2021. (Acceptance rate: 18.8%)
- [26] Michael Wang, **Chenglong Fu**, Xiaojiang Du. “Decision-Tree Based Root Cause Localization for Anomalies in Smart IoT Systems.” In the IEEE International Conference on Communications (ICC), June 2021
- [27] Heena Rathore, **Chenglong Fu**, Amr Mohamed, Abdulla Al-Ali, Xiaojiang Du, Mohsen Guizani, and Zhengtao Yu. “Multi-layer security scheme for implantable medical devices.“ Neural Computing and Applications 32, no. 9 (2020): 4347-4360.
- [28] Xuening Xu, **Chenglong Fu**, Xiaojiang Du, and E. Paul Ratazzi. “Effective UAV and Ground Sensor Authentication.” In IEEE Global Communications Conference (GLOBECOM), December 2019.
- [29] **Chenglong Fu**, Xiaojiang Du, Longfei Wu, Qiang Zeng, Amr Mohamed, and Mohsen Guizani. “POKs-based secure and energy-efficient access control for implantable medical devices.“ In International Conference on Security and Privacy in Communication Systems (SecureComm), October 2019. (Acceptance rate: 25.5%)
- [30] Qiang Zeng, Jianhai Su, **Chenglong Fu**, Golam Kayas, Lannan Luo, Xiaojiang Du, Chiu C. Tan, and Jie Wu. “A multiversion programming inspired approach to detecting audio adversarial examples.” In 49th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), June 2019. (Acceptance rate: 21%)
- [31] **Chenglong Fu**, Qiang Zeng, and Xiaojiang Du. “Towards Efficient Integration of Blockchain for IoT Security: The Case Study of IoT Remote Access.” arXiv preprint, 2019.
- [32] **Chenglong Fu**, Tassadit Kezmane, Xiaojiang Du, Yat Fu, and Colin Morrisseau. “An location-aware authentication scheme for cross-domain internet of thing systems.” In International Conference on Computing, Networking and Communications (ICNC), March 2018.

FUNDINGS AND GRANTS

Lead PI - UNC Charlotte Core Facilities Program

Sep 2024

- “Building Customizable Large Language Model Services for Education and Research”, Granted by University of North Carolina at Charlotte, \$250K. 2024-2026

CoPI - NC Research Opportunities Initiative

June 2023

- “North Carolina Research Capacity Building for a Secure & Reliable Power Grid 2050”, Granted by University of North Carolina System, \$1.5M. 2023-2026

PI - UNCC Faculty Research Grants Program

Jan 2024

- “Automated Auditing of Initial Coin Offerings”, Granted by the University of North Carolina at Charlotte, \$8K. 2024-2025

TEACHING EXPERIENCES

ITIS-3200 (Introduction to Security and Privacy) - UNC Charlotte

Jan 2025 – May 2025

ITIS-4221/5221 (Penetration Testing & Secure Programming) - UNC Charlotte

Jan 2023 – Present

- Class Evaluation: 4.0/5

ITSC-3146 (Operating System & Computer Networks) - UNC Charlotte

Jan 2023 – May 2023

- Class Evaluation: 4.1/5

PROFESSIONAL SERVICES

Conference organization

- IEEE/ACM International Symposium on Quality of Service 2023, Publicity Chair
- IEEE/ACM IEEE International Conference on Cloud Networking 2023, Publicity Chair
- IEEE International Conference on Intelligence and Security Informatics 2023, Finance Chair

Reviewers

- IEEE Transactions on Information Forensics and Security, Reviewer
- IEEE International Conference on Computer Communications 2021-2023, Reviewer

AWARDS AND HONORS

Scott Hibbs Future of Computing Award

April 2021

- Each year, only one student is selected from all the graduate and undergraduate students (over 1,000 students) in the CIS Department at Temple University

Outstanding Undergraduate Scholarship

Sep 2014

- 10% among all undergraduates, granted by University of Science & Technology of China

Seagate Scholarship

Dec 2013

- 5% among all undergraduates, Granted by Seagate Technology co., LTD

Outstanding College graduates in Provincial Scale

June 2015

- 1 among all undergraduates in the department of Information Security, Granted by Department of Education of Anhui Province